

Some latest wireless technology applications

Technology	Features	Advantages	Disadvantages
BeamCaster	A local wireless network, BeamCaster is the latest Li-Fi technology that transfers information to electronic devices with the help of light. BeamCaster has a router at its core that is capable of transmitting signal to a range of about 7-8 metres with the help of a light beam. It is about four times faster in speed than Wi-Fi standards and can transmit signals to around eight devices at a time.	➤ Has eight receivers for multiple connections. ➤ Data transfer speed is 1.25Gbps, guaranteed to each working unit. ➤ Signal transmission range of 7 to 8 metres. ➤ Converts data into digital binary code during transfer through a light beam, so it is impossible to either intercept it or to eavesdrop on it.	➤ Difficulty in positioning the receivers. ➤ Inconvenience in working with mobile devices. ➤ High implementation cost relative to other wireless networks.
Twisted radio beams	A new way of wireless communication through radio waves is through twisted radio beams. Several polarised radio beams are twisted together into a spiralled beam that allows data to be sent at a speed of 32Gbps across 2.5 metres in open air. Twisting the beams is basically a form of orbital angular momentum (OAM) multiplexing.	➤ Sends data at a rate of 32Gbps. ➤ Can attain high data rate through OAM multiplexing. OAM allows data to be transferred through multiple channels at a single frequency. ➤ High data capacity due to increased surface area that results due to twisting of beams.	➤ Maintains NFC at low frequency. For an efficient OAM based communication, near field communication will have to be maintained at low frequencies to force the antennae to rapidly grow in size as the distance grows. This would reduce interference to a larger extent.
IPCF-MC	Industrial Point Coordination Function-Management Channel (IPCF-MC) is an industry-specific enhancement to wireless local area network technology (WLAN). It is optimised for applications with freely movable wireless network nodes, like Simatic Mobile Panel and guided vehicle system, and focuses on optimising the roaming process to redirect the freely-movable wireless network nodes.	➤ Provides dual access points to broadcast administrative frames. ➤ Two processes go on at the same time—first, data exchange between client module and access point; second, radio interface of each access point transmits on the same radio channel, allowing simultaneous exchange of data. ➤ Single failure point that is the client module. ➤ Selective transmission results in increased range, transmitting high power.	➤ No dual IWLAN dual client feature.

Vehicles communication enhanced from Honda to Railways

Security has been an important factor when it comes to vehicles. Its not only the security inside the vehicle but also the security outside it. The dedicated short range communications (DSRC) technology is helping this security to go to the next level. It is part of the cooperative intelligent transport systems (C-ITS). It is a two-way short-to-medium-range wireless communication capability that permits very high data transmission critical in communications based active safety applications.

Honda is enabling DSRC in its vehicles to enable advanced vehicle-to-pedestrian (V2P) and vehicle-to-motorcycle (V2M) technologies to support on-road wireless communication. It is using DSRC wireless technology, along with the pedestrian's smartphone GPS and its dynamic sensing capability in the 5.9GHz band, to establish a communication channel between the pedestrian's smartphone and nearby vehicles to determine if the pedestrian is in danger of being struck by an oncoming car. The V2P system is effective even when the pedestrian is not easily detectable by the driver, like in conditions when stepping off a curb from behind a parked vehicle or some other traffic obstruction. Whereas, V2M detects the presence of a motorcycle even when it is obstructed from the view of nearby automobile drivers.

On the other hand, Railways is implementing this technology to provide three levels of warning and avoid any kind of accidents. The first of these warning is to inform that there is a crossing ahead, the second is to inform that there is a train in the vicinity and the final one is to inform about a collision and requirement of an emergency action.

DSRC is for data-only systems and operates on radio frequencies in the 5725MHz to 5875MHz industrial, scientific and medical (ISM) band. DSRC systems consist of road side units and the on-board units with transceivers and transponders. The DSRC standards specify the operational frequencies and system bandwidths, but also allow for optional frequencies that are covered (within Europe) by national regulations.

Although this technology has been there for quiet some time, it has been only a few years since its real implementation started.

challenge with this 802.16n standard is that, although it provides multicast communication with MKM, it does not support the re-keying of multicast authentication key.

While going through the new standards, you would have come across the advancements to the technologies as well as the various challenges. Despite these challenges, you will realise that Bluetooth 4.1 with Bluetooth Smart is the most popular technology in the industry today because of the wide acceptability of Bluetooth amongst mobile users and its less-interference feature. Bluetooth has evolved a lot over a period of time, from a high-energy, simple Bluetooth network for limited applications to a low-energy Bluetooth Smart for a large number of applications.

However, the next challenge in this era of new wireless technologies is Wi-Fi with gigabytes-speed of communication. Whereas, technologies like RF based ZigBee Remote Control 2.0 or the multicast key management with IEEE 802.16n are getting more and more reliable to outcast themselves in the competition. ●

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and non-HR stations are out of the scope of IEEE 802.16n and are compatible only with IEEE 802.16-2009/802.16j or IEEE 802.16m. The other